

CHM 120

Chromatography of Food Dyes

Final Report

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Test Your Knowledge

Categorize the following statements as true or false.

True	False
1 Chromatography is a technique commonly used to separate mixtures of chemical compounds into their constituent components. The leading edge of the solvent, as it moves up the chromatography paper, is called the solvent front.	2 The choice of a solvent in paper chromatography is unimportant. An adsorbent is a solid that lacks affinity for the solvent and the chemical components of the mixture.

Match each type of chromatography with the best description.

Solid particles are spread over a flat glass or plastic surface and a solvent is allowed to travel up through the solid particles	1	Paper
Solvent travels over a column of tiny beads	2	Liquid
Vapor travels over a column of tiny beads	3	Gas
A sheet of cellulose is placed in a liquid, which travels up the sheet	4	Thin-layer

Match the R_f value to the appropriate paper chromatography results.

R_f Value		
	Solvent distance = 68 mm, Dye distance = 31 mm	1 $R_f = 0.46$
	Solvent distance = 64 mm, Dye distance = 58 mm	2 $R_f = 0.91$

Exploration

The two phases of chromatography are the ____.

- ☒ solid phase and mobile phase
- ☐ solid phase and soluble phase
- ☐ stationary phase and mobile phase
- ☐ stationary phase and column phase

Liquid and gas chromatography are also called ____ chromatography due to the equipment design.

- ☒ column
- ☐ particle
- ☐ adsorbent
- ☐ eluent

Chromatogram results are considered to have better resolution if the R_f values of the components being separated are farther apart rather than closer together.

- ☒ True
- ☐ False

FD&C dyes are specific dyes banned from human consumption under the Food, Drugs and Cosmetics Act.

- ☐ True
- ☒ False

R_f values are calculated by dividing the ____ by the ____.

- ☐ solvent front distance; center of compound distance from base line
- ☒ distance the compound moved; solvent front distance
- ☐ length of compound spot; width of compound spot
- ☐ number of compound spots; mass of solution

Exercise 1

What might happen if a student used a pen to mark the baseline on the chromatography paper?

B / U

29 Word(s)

The ink from the pen may interfere with the lab. It may effect the dyes. And if it gets messed up while being drawn, it isn't easy to erase.

What would happen if the student did not stop the development before the solvent reached the top of the paper?

B / U

22 Word(s)

The water will soak all the way up to the top. It can have an effect on the formation of the spots

What would happen if the solvent level was above that of the dye spots on the paper at the start of the development?

B / U

18 Word(s)

It would have been too much for the paper. It would all be wet, and probably somewhat fast..

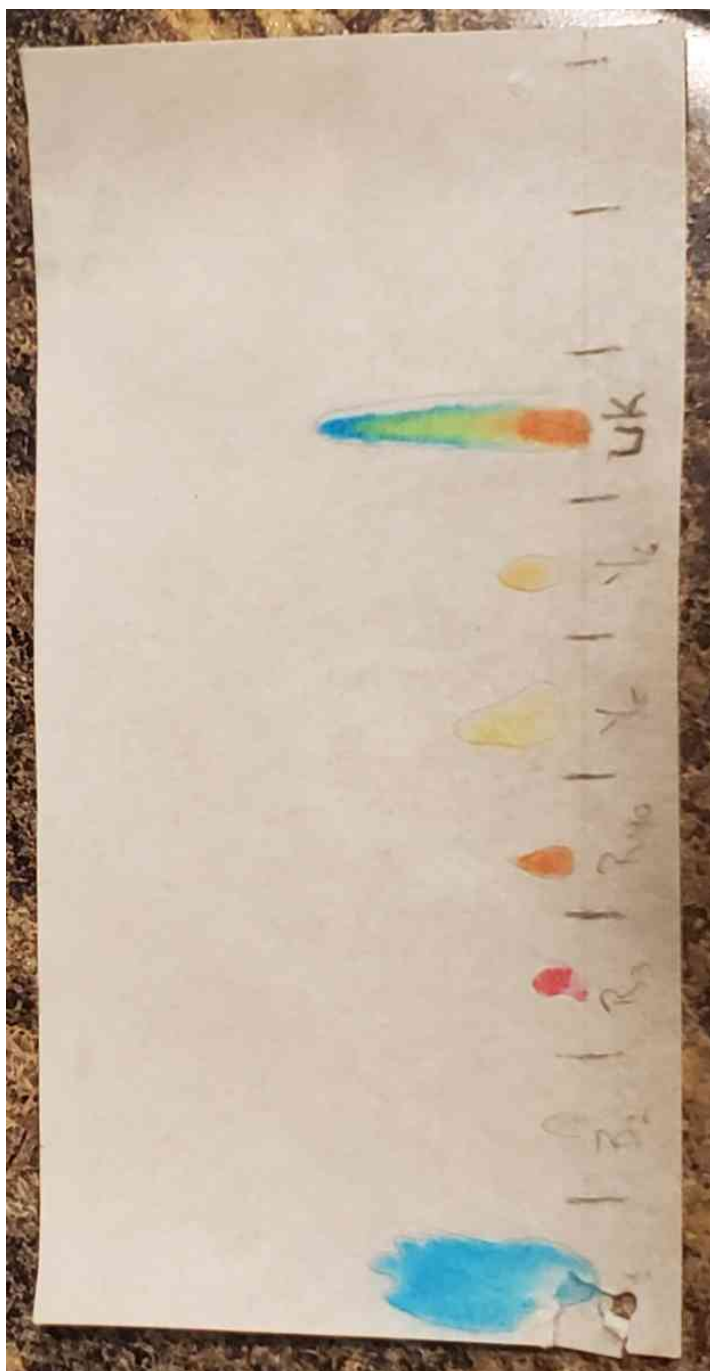
Why is it important to mark the solvent level on the chromatography paper as soon as you remove it from the Petri dish?

B / U

20 Word(s)

Because the paper starts to dry, and once it is dry you cannot see where the solvent level reached to.

Photo 1: Chromatogram



Data Table 1: Chromatogram Data

Sample	Solvent front (mm)
Blue 1	55
Blue 2	55
Red 3	55
Red 40	55
Yellow 5	55
Yellow 6	55
UK	55

Sample	Spot 1 mm
Blue 1	11
Blue 2	3
Red 3	3
Red 40	4
Yellow 5	9
Yellow 6	7
UK	7

Sample	Spot 1 R _f
Blue 1	.2
Blue 2	.05
Red 3	.05
Red 40	.07
Yellow 5	.16
Yellow 6	.13

UK	.13
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Sample	Spot 2 mm
Blue 1	
Blue 2	
Red 3	
Red 40	
Yellow 5	
Yellow 6	
UK	13

Sample	Spot 2 R _f
Blue 1	
Blue 2	
Red 3	
Red 40	
Yellow 5	
Yellow 6	
UK	.24

Sample	Spot 3 mm
Blue 1	
Blue 2	
Red 3	
Red 40	
Yellow 5	

Yellow 6	
UK	15
Sample	Spot 3 R _f
Blue 1	
Blue 2	
Red 3	
Red 40	
Yellow 5	
Yellow 6	
UK	.27

Data Table 2: FD&C Dyes in Sample

Sample:	FD&C Dyes in Sample
Unknown Sample	Yellow 6, Blue 1, Possibly Yellow 5, Even though it didn't seem close based on R _f values, I'd argue Red 40 was also used

Competency Review

Chromatography is useful for identifying unknown substances in a ____.

- ☐ pure solution
- ☒ mixture
- ☐ solid

A(n) ____, used as part of the stationary phase, has an affinity for the solvent and the chemical components of the mixture.

- ☐ adsorbent
- ☒ eluent
- ☐ Rf factor
- ☐ chromatogram

____ chromatography uses a column of solid beads and a liquid solvent to separate mixtures.

- ☐ Paper
- ☐ Thin-layer
- ☐ Gas
- ☒ Liquid

In paper chromatography, a strip of paper is utilized as the ____.

- ☐ mobile phase
- ☒ stationary phase
- ☐ solvent front
- ☐ eluent

R_f values of unknown compounds are compared to standards to determine their identity.

- ☒ True
- ☐ False

FD&C dyes are used to color food items with hues of ____.

- ☐ blue
- ☐ yellow
- ☐ red
- ☒ All of the above

During a paper chromatography experiment using food dyes, saltwater serves as the ____ in the experiment.

- ☐ adsorbent
- ☒ eluent
- ☐ stationary phase
- ☐ unknown component

During paper chromatography experiments, why is the solvent front marked with a pencil as soon as the filter paper is removed from the chamber?

- ☐ To identify the location of the solvent front once the paper dried
 - ☐ To measure the distance traveled by the solvent during the development
 - ☐ To determine the retention factor of the compound
 - ☒ All of the above
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An unknown food dye produced two spots on a chromatogram with R_f values of 0.15 and 0.23. Use the table below to identify the unknown compounds.

Sample	Solvent front (mm)	Spot 1 (mm)	Spot 1 R_f
Blue 1	60	40	0.67
Blue 2	60	6	0.1
Red 3	60	4	0.07
Red 40	60	10	0.17
Yellow 5	60	23	0.38
Yellow 6	60	13	0.22

- ☐ Blue 1 and Red 3
- ☐ Yellow 5 and Red 40
- ☐ Blue 2 and Yellow 6
- ☒ Red 40 and Yellow 6

Extension Questions

Metal ions can be separated by paper chromatography through a process similar to the one you used to separate food dyes. However, the metal ions are not as distinctly colored as the food dyes, so often an additional step is needed where the developed paper is treated with a chemical agent that will make the metal ions appear visible.

The table below gives some data that was obtained running both some known metal ions and an unknown sample containing one or more metal ions.

Sample	Color of Treated Component	Distance Component Moved (mm)	Distance Solvent Moved (mm)
Cu ⁺² ion	Blue	45	60
Ni ⁺² ion	Brown	30	60
Co ⁺² ion	Brown	18	60
Fe ⁺² ion	Red	55	60
Unknown	Brown	20	60
	Blue	43	

a. What ion(s) is/are contained in the unknown solution?

b. Which of the metal ions has the strongest affinity for the solvent system?

c. If the solvent was allowed to travel 80 mm instead of the 60 mm used in this experiment, at what distance would you expect to find the Ni ion located?

(a)

Cu⁺²
Co⁺²

(b)

Co⁺²

(c)

Distance of Ni ion should be roughly 40mm if solvent traveled 80mm

